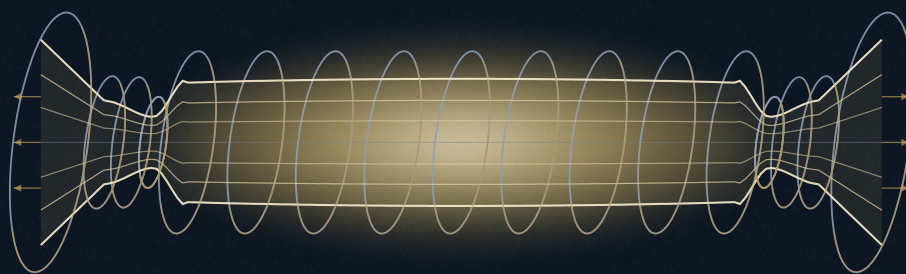




KRONOS FUSION ENERGY



PAPER 2.27 OF THE 2026 KRONOS SET

The Advanced-Fuel Scorecard

Physics, Not the Aneutronic Ideal

Grading D-T, D-D, D-³He, and p-¹¹B on one ruler — and letting the Kronos architecture fall out of the grades.

D-T breeding · D-³He power (f_n 5.44%) · p-¹¹B not chosen

Explore it live — [3-D model](#) · [physics validator](#) · [film series](#)

THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

The Advanced-Fuel Scorecard — Physics, Not the Aneutronic Ideal

Fuel choice is the most consequential decision in a fusion program, and it is often made emotionally — toward the aneutronic ideal — rather than on the physics. We grade the four candidate fuels on one ruler: reactivity, temperature, neutron fraction, tritium logistics, availability, and demonstrated gain. No fuel wins on every axis. The Kronos architecture follows the scorecard, not the ideal: run deuterium–tritium where the product is the neutron and the bred material (the breeder), and deuterium–helium-3 (only 5.44% neutronic) where the product is power and neutrons are the liability (the burner) — with the helium-3 supply honestly lunar-gated.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132158](https://doi.org/10.5281/zenodo.22132158) (v2 [10.5281/zenodo.21746479](https://doi.org/10.5281/zenodo.21746479)) · Zenodo community *kronos_fusion_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

Keywords:

advanced fusion fuels

deuterium–helium-3

proton–boron

neutron fraction

fuel selection

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
$FOAK/NOAK/BOAK$	first / n th / best of a kind
<i>Hyperion</i>	the ST breeder (internal: Mode D2)
<i>Aegis</i>	the generator, defense/installation housing (internal: Mode M)
<i>MetroVolt</i>	the generator, data-center housing (Mode M)

§ 1

Fuel choice is the decision

Every fusion company implicitly picks a fuel, and the pick determines everything downstream: the neutron load, the tritium logistics, the required temperature, the conversion strategy. The choice is too often made emotionally — toward the aneutronic ideal — rather than on the physics. We grade all four candidate fuels on the same ruler and let the Kronos architecture fall out of the grades.

THE RESULT IN ONE LINE

No fuel wins on every axis. Kronos runs **deuterium–tritium where the product is the neutron and the bred material** (the breeder) and **deuterium–helium-3 where the product is power** and neutrons are the liability (the burner, only 5.44% neutronic) — with helium-3 honestly lunar-gated. Proton–boron is *not* chosen.

The scorecard

The single most decisive axis is the neutron fraction of the fusion power — it sets the shielding, the activation, and whether the machine can sit near people and hardware. Deuterium–tritium is ~80% neutronic; deuterium–deuterium ~66%; deuterium–helium-3 only 5.44%, and even those neutrons come from an unavoidable D–D side reaction; proton–boron is effectively aneutronic at under 1%.

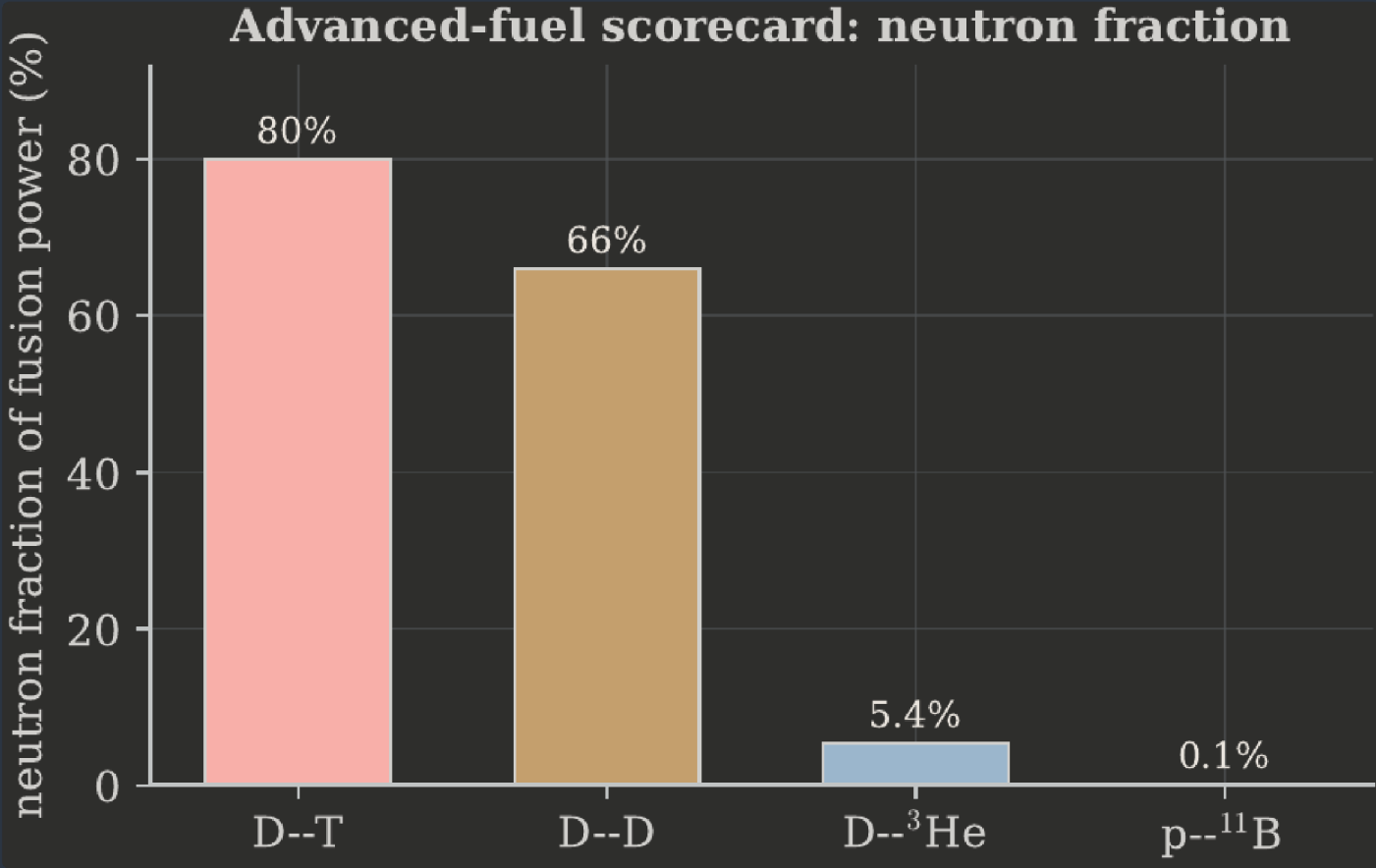


FIGURE 1. The decisive axis: neutron fraction of fusion power across the four fuels. Deuterium–helium-3 combines a low neutron burden with accessible reactivity. **PUBLIC REACTION PHYSICS**

D–T	Highest reactivity, lowest temperature · but ~80% neutronic and consumes tritium · <i>demonstrated net gain (NIF)</i>
D–D	No tritium needed · but ~66% neutronic and low reactivity
D– ³ He	Only 5.44% neutronic, reactivity still accessible · but requires helium-3 Earth lacks in bulk
P– ¹¹ B	Effectively aneutronic · but highest temperature, radiates hardest, <i>no device has reached net gain</i>

The Kronos reading of the scorecard

The architecture follows the grades rather than the ideal. Where the product is the *neutron and the bred material* — tritium, helium-3, isotopes, materials qualification — deuterium–tritium is correct, because its high neutron output is the whole point; this is the breeder. Where the product is *power* and neutrons are a liability to shield and a source of activation, deuterium–helium-3 is correct, because 5.44% is the lowest neutron burden among fuels with accessible reactivity; this is the burner.

Proton–boron is not chosen: its temperature, its radiation, and the absence of any demonstrated gain outweigh its aneutronic appeal *today*. And the one honest cost of the deuterium–helium-3 choice — fuel availability — is met by the breeder’s co-product helium-3 and, at fleet scale, lunar supply gated around 2038–40. The scorecard, not the aneutronic ideal, drew the architecture. DESIGN RATIONALE

A P P A R A T U S

Portfolio, People & Record

*The intellectual-property estate, the people who built it, the limitations
that gate it, and the sources it rests on.*

1

GRANTED PATENT

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2026 PROVISIONALS

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INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

FOUNDING PARTNERS — SCIENCE

Dr. Gerald Kulcinski

Co-Founder · Helium-3 & Advanced-Fuel Fusion
UW-Madison · 2023–Present

Dr. Carl Weggel

Chief Scientist / S.M.A.R.T. Design
MIT Alcator Program · 2022–Present

Dr. Robert J. Weggel

Magnetic Field Design
MIT Magnet Lab · 2022–Present

BOARD

Priyanca Ford

Founder & CEO · Executive Board
2022–Present

Bandel Carano

Finance / Strategy
Oak Investment Partners / Stanford · 2025–2026

Patrick Schweiger

Fusion Device Engineering
Oklo / CFS / TerraPower · 2025–Present

SCIENTIFIC ADVISORS

Dr. Steven O. Dean

Fusion Policy & History
DOE / Fusion Power Associates · 2025–Present

Dr. Patrick H. Diamond

Plasma Turbulence & Transport
UC San Diego · 2025–Present

Dr. Jack J. Dongarra

HPC & Numerical Algorithms
Turing Award · 2025–Present

Dr. Nasr M. Ghoniem

Chief Material Scientist
UCLA · 2025–Present

Dr. Siegfried Glenzer

Plasma Physics & Laser Fusion
Stanford / SLAC · 2022–Present

Dr. David A. Hammer

Pulsed-Power Fusion
Cornell · 2025–Present

Dr. Donald A. Spong

Plasma Theory & Confinement
ORNL · 2025–Present

Dr. Curtis Smith

Risk Assessment & Nuclear PRA
MIT / Idaho National Lab · 2025–Present

RADM (Ret.) David Goggins

Naval Engineering & Power-Plant Design
U.S. Navy · 2023–2026

Dr. Konstantin Batygin

Mathematician
Caltech · 2022–2025

Dr. Ruben Fair

Magnet Design / ITER Liaison
PPPL / Jefferson Lab · 2022–2025

Dr. Paul S. Weiss

Materials & Nanotechnology
UCLA · 2022–2025

SCIENTIFIC AUDITORS**Dr. Wilfred A. Cooper**

Plasma Equilibrium Theory
EPFL / Swiss Plasma Center · 2025–Present

Dr. Ahmed Hassanein

Plasma–Material Interactions
Purdue / Argonne · 2025–Present

Dr. Patrick E. Hopkins

Extreme Thermal Materials
U. of Virginia · 2025–Present

Dr. Peter Hosemann

Materials Joining & Structural Integrity
UC Berkeley · 2025–Present

Dr. Travis W. Knight

Advanced Nuclear Fuels & Systems
U. of South Carolina · 2025–Present

Dr. Philippe Lebrun

Cryogenics & Magnet Cooling
CERN · 2025–Present

Dr. Nitendra Singh

Nuclear Safety & Fuel Cycle
ITER · 2025–Present

Dr. Guido Van Oost

Magnetic Confinement & Fusion Systems
Ghent University · 2025–Present

Dr. Gary S. Was

Radiation Materials & Structural Integrity
U. of Michigan · 2025–Present

Dr. Ray Sedwick

Direct Energy Conversion
U. of Maryland · 2025

OPERATIONS**Michael Laughlin**

Chief Operating Officer
2022–Present

Martin Owens

Chief Strategy Officer
LANL / GE Hitachi · 2022–Present

MG (Ret.) Paul Pardew

Chief Contracting Officer
Army Contracting Command · 2022–Present

David Beck

Fusion Commercialization
U.S. Space Force · 2024–Present

Sushma Bhatia

Environmental & Legislative
Google · 2022–Present

Michael De Frenza

Technology Licensing
2023–Present

MG (Ret.) Robin L. Fontes

Cybersecurity & Defense
Army Cyber Command · 2022–Present

Vijay Gehani

Supply Chain & Components
INOX India · 2024–Present

Jon Michel Greenwood

IT & AI Infrastructure
Live Nation · 2023–Present

Brian C. O'Neill

National Security
CIA / ODNI · 2022–Present

Gen. (Ret.) Gustave F. Perna

DoD Liaison
Operation Warp Speed · 2022–2023

Andrea Romero

Marketing Lead
2022–Present

Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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REFERENCES

Sources

- [1] H.-S. Bosch and G. M. Hale, “Improved formulas for fusion cross-sections and thermal reactivities,” *Nuclear Fusion* **32**, 611 (1992).
- [2] A. S. Richardson, *NRL Plasma Formulary*, Naval Research Laboratory (2019).
- [3] I. E. Ochs, E. J. Kolmes, and N. J. Fisch, “Bremsstrahlung and the electron–electron contribution in fusion plasmas,” *Physics of Plasmas* (2024).
- [4] T. H. Rider, “Fundamental limitations on plasma fusion systems not in thermodynamic equilibrium,” *Physics of Plasmas* **4**, 1039 (1997).
- [5] ITER Physics Expert Groups, “ITER Physics Basis,” *Nuclear Fusion* **39**, 2137 (1999).
- [6] N. A. Uckan and J. Sheffield, “Tokamak scaling and confinement,” in *Tokamak Concept Improvement*, IAEA (1989).
- [7] O. Sauter, C. Angioni, and Y. R. Lin-Liu, “Neoclassical conductivity and bootstrap current,” *Physics of Plasmas* **6**, 2834 (1999).
- [8] F. Troyon *et al.*, “MHD limits to plasma confinement,” *Plasma Physics and Controlled Fusion* **26**, 209 (1984).
- [9] J. P. Freidberg, *Plasma Physics and Fusion Energy*, Cambridge University Press (2007).
- [10] R. F. Post, “The magnetic mirror approach to fusion,” *Nuclear Fusion* **27**, 1579 (1987).
- [11] D. E. Baldwin, “End-loss processes from mirror machines,” *Reviews of Modern Physics* **49**, 317 (1977).
- [12] T. K. Fowler, R. W. Moir, and T. C. Simonen, “A new simpler tandem mirror,” *Nuclear Fusion* **57**, 056014 (2017).
- [13] G. L. Kulcinski and J. F. Santarius, “Advanced fuels and the D–³He cycle,” University of Wisconsin Fusion Technology Institute.
- [14] W. L. Barr and R. W. Moir, “Test results on direct energy converters for mirror reactors,” *Nuclear Technology/Fusion* **3**, 98 (1983).
- [15] D. B. Go *et al.*, “Thermionic energy conversion in the twenty-first century,” *Frontiers in Mechanical Engineering* **3**, 13 (2017).
- [16] M. Wang *et al.*, “The AME 2020 atomic mass evaluation,” *Chinese Physics C* **45**, 030003 (2021).
- [17] J. E. Menard *et al.*, “Fusion nuclear science facilities and pilot plants based on the spherical tokamak,” *Nuclear Fusion* **56**, 106023 (2016).
- [18] A. J. Creely *et al.*, “Overview of the SPARC tokamak,” *Journal of Plasma Physics* **86**, 865860502 (2020).
- [19] T. P. Wright, “Factors affecting the cost of airplanes,” *Journal of the Aeronautical Sciences* **3**, 122 (1936).
- [20] U.S. Department of Energy / NNSA, *Stockpile Stewardship and Management Plan*, FY2024.
- [21] J. F. Santarius, G. L. Kulcinski, and L. A. El-Guebaly, “A strategy for D–³He fusion energy development,” *Journal of Fusion Energy* **17**, 33 (1998).
- [22] M. A. Abdou *et al.*, APEX advanced blanket concepts, *Fusion Engineering and Design* (1999–2003).
- [23] Federal Register, notice on supply of tritium to non-federal entities (1999).
- [24] Kronos Fusion Energy, “Simulation register and reproducibility deposit,” Zenodo (2026).
- [25] A. O. Thome *et al.*, “ELM-free negative-triangularity edge on DIII-D,” *Plasma Physics and Controlled Fusion* (2024), doi:10.1088/1361-6587/ad6f40.
- [26] T. Eich *et al.*, “Scaling of the tokamak near scrape-off layer power width,” *Nuclear Fusion* **53**, 093031 (2013).
- [27] J. Lim *et al.*, “Scrape-off-layer width in negative-triangularity plasmas,” *Nuclear Fusion / Plasma Physics and Controlled Fusion* (2023).

[28] A. O. Nelson *et al.*, “Prohibition of second-stability access at negative triangularity,” *Plasma Physics and Controlled Fusion* **64**, 124002 (2022).

COLOPHON

How this document was made

The Kronos Fleet — Combined Editorial, 2026 is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

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Numbers and their classes. Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

Public edition. This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

PAPER I	Hyperion — The Tritium Foundry · 10.5281/zenodo.21746157
PAPER II	Aegis & MetroVolt — The Generator · 10.5281/zenodo.21746479
PAPER III	High-Field REBCO Magnets & Tape · 10.5281/zenodo.21842514
PAPER IV	Direct Energy Conversion · 10.5281/zenodo.21842864
PAPER V	AI, ML & Quantum Control · 10.5281/zenodo.21842371

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KRONOS FUSION ENERGY

The Advanced-Fuel Scorecard

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